KAPREKAR CONTEST – FINAL – SUB-JUNIOR

(Classes VII and VIII)

AMTI – NOVEMBER, 2018.

Instructions:
1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Fill in FACE SLIP and your rough working should be in the answer paper itself.
5. Maximum time allowed is THREE hours.
6. All questions carry equal marks.

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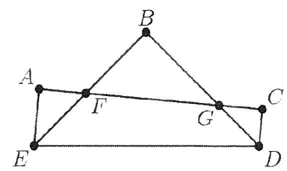
1. A lucky year is one in which at least one date, when written in the form day / month / year, has the following property: The product of the month times the day equals the last two digits of the year. For example, 1956 is a lucky year because it has the date 7 / 8 / 56 where $7 \times 8 = 56$, but 1962 is not a lucky year as $62 = 62 \times 1$ or 31×2, where 31/2/1962 is not a valid date. From 1900 to 2018 how many years are not lucky (not including 1900 and 2018)? Give proper explanation for your answer.

2. In the figure given, $\angle A$, $\angle B$ and $\angle C$ are right angles. If and $\angle AEB = 40°$ and $\angle BED = \angle BDE$, then find $\angle CDE$.



3. (a) $ABCDEF$ is a hexagon in which $AB = BC = CD = DE = 2$ and $EF = FA = 1$. Its interior angle C is between 90° and 180° and F is greater than 180°. The rest of the angles are 90° each. What is its area?

   (b) A convex polygon with 'n' sides has all angles equal to 150°, except one angle. List all possible values of n.

4. a, b, c are distinct non-zero reals such that $\frac{1+a^3}{a} = \frac{1+b^3}{b} = \frac{1+c^3}{c}$. Find all possible values of $a^3 + b^3 + c^3$.

5. Find the smallest positive integer such that it has exactly 100 different positive integer divisors including 1 and the number itself.

6. (a) What is the sum of the digits of the smallest positive integer which is divisible by 99 and has all of its digits equal to 2?

   (b) When 270 is divided by the odd number n, the quotient is a prime number and the remainder is 0. What is n ?

7. Consider the sums

$$A = \frac{1}{1\cdot 2} + \frac{1}{3\cdot 4} + \ldots + \frac{1}{99\cdot 100} \quad and \quad B = \frac{1}{51\cdot 100} + \frac{1}{52\cdot 99} + \ldots + \frac{1}{100\cdot 51}.$$

   Express $\frac{A}{B}$ as an irreducible fraction.

8. Let a, b, c be real numbers, not all of them are equal. Prove that if $a+b+c=0$ then $a^2+ab+b^2 = b^2+bc+c^2 = c^2+ca+a^2$.

   Prove the converse, if $a^2+ab+b^2 = b^2+bc+c^2 = c^2+ca+a^2$ then $a+b+c=0$.
